

Week 5

System Heterogeneity, Client Selection, and Communication Efficiency

Federated learning as a system problem

Graduate course · English instruction



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Optimization Is Only Half the Problem

A mathematically strong FL algorithm can still be impractical when clients are slow, unreliable, or expensive to communicate with.

Compute

CPU/GPU speed, memory limits, and local training time.

Network

Bandwidth, latency, packet loss, and uplink bottlenecks.

Availability

Charging state, Wi-Fi, idleness, dropout, and disconnection.

Good optimization



system constraints



real wall-clock behavior

In FL, systems decisions can change both speed and the statistical sample that reaches the server.

Learning Objectives

Separate heterogeneity

Distinguish statistical heterogeneity from hardware, network, energy, and availability constraints.

Explain stragglers

Describe why synchronous rounds can wait for the slowest required client.

Compare client selection

Reason about random, resource-aware, utility-aware, and diversity-aware strategies.

Design communication efficiency

Use local updates, quantization, sparsification, pruning, and partial updates appropriately.

By the end, you should be able to design an FL round under realistic resource constraints—and defend the trade-offs.

Statistical vs. System Heterogeneity

Statistical heterogeneity

Clients differ in what data they observe.

Labels

Different class proportions

Features

Different sensors, locations, users

Quantity

Different dataset sizes

System heterogeneity

Clients differ in how they participate.

Compute

Different training speed

Network

Different bandwidth / latency

Availability

Different energy / uptime

The two can interact: excluding slow clients may also exclude a specific data population.

The Edge-Client Constraint Set

Computation

Different processors → different local training time

Memory

Large models may not fit on every device

Energy

Battery-powered clients cannot train indefinitely

Bandwidth

Uplink can be the dominant bottleneck

Latency

Network delay changes round duration

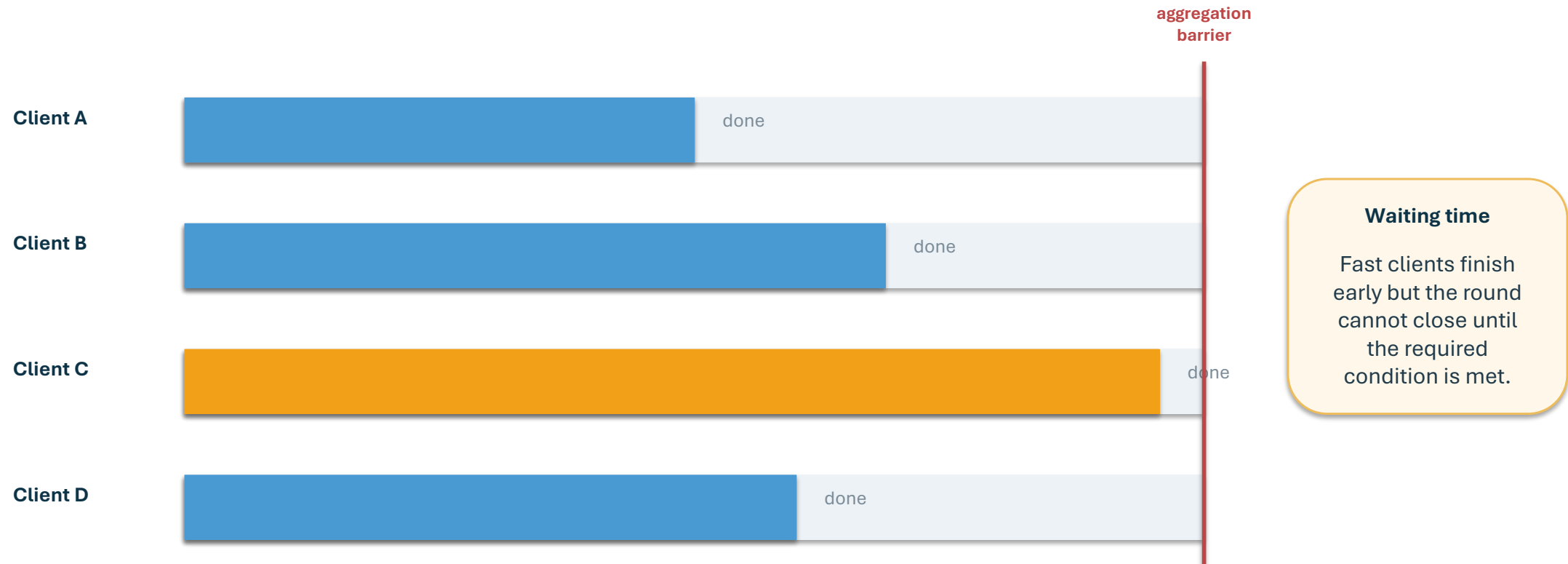
Availability

Eligibility depends on charging, Wi-Fi, idleness, and user activity

Participation is state-dependent: a client can be useful statistically but unavailable operationally.

The Straggler Problem

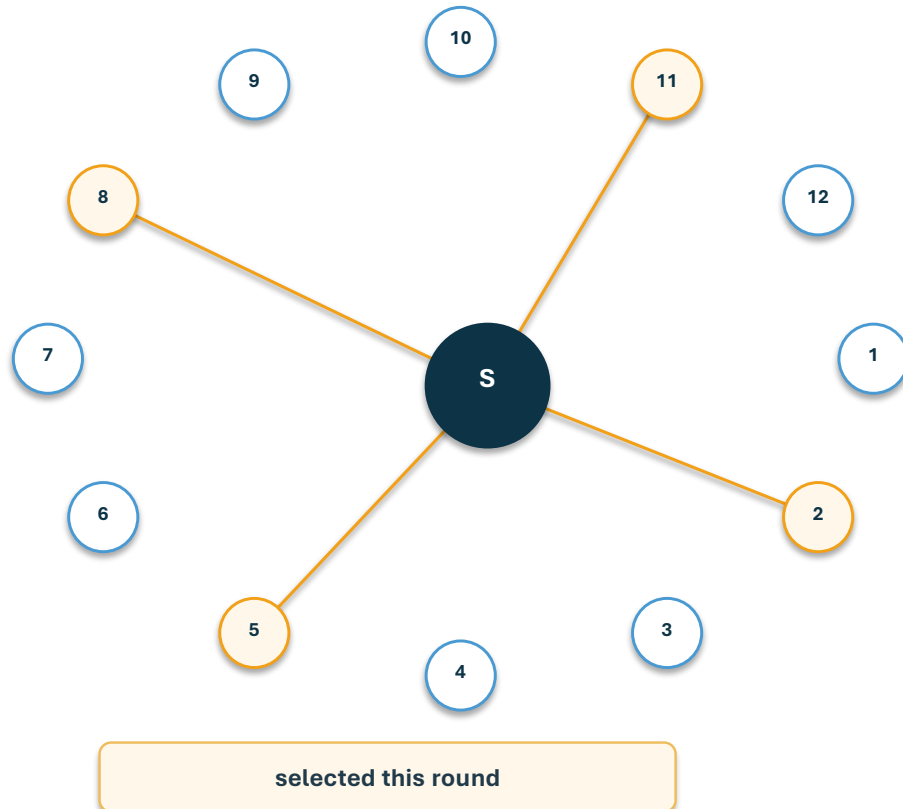
Synchronous aggregation creates a barrier



Dropping stragglers lowers latency—but may bias training if slow clients are statistically different.

Partial Client Participation

Large populations make all-client participation unrealistic.



Sampling variance

A different subset contributes in each round.

Bias risk

Eligible or fast clients may not represent the full population.

Dropout handling

Some selected clients may fail before upload.

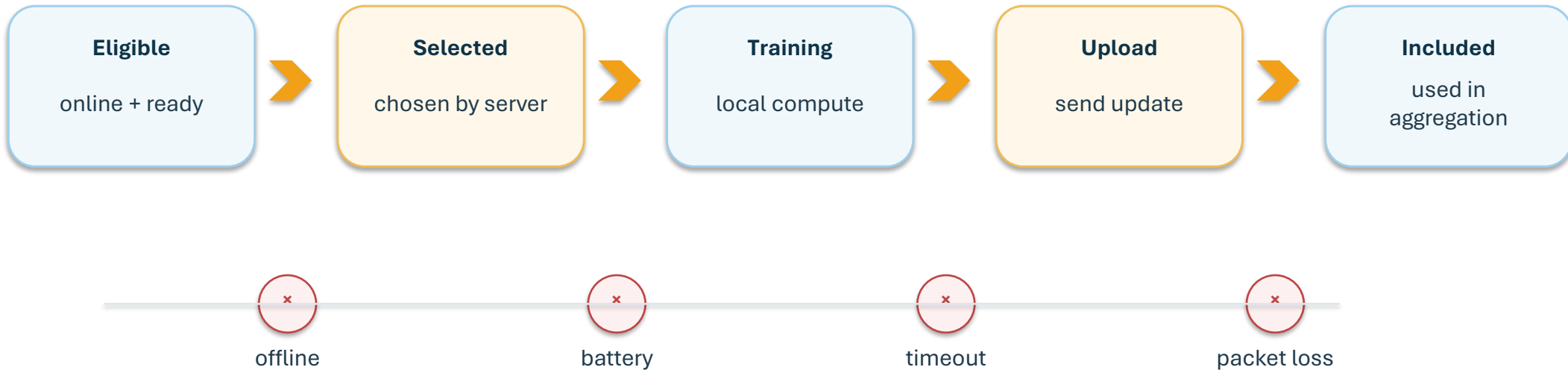
Scalability

Partial participation makes large FL systems feasible.

Partial participation is necessary—but it turns client sampling into part of the learning algorithm.

Client Availability and Dropout

A client can fail at several points in the round.



Design implication

Eligibility rules, deadlines, retries, and aggregation thresholds must be explicit.

Statistical implication

Dropout is harmless only if missing clients are effectively random; otherwise it can create bias.

Client Selection Strategies

Strategy	Selection signal	Potential benefit	Risk
Uniform random	Availability only	Simple and statistically clean	May waste scarce resources
Resource-aware	Bandwidth, compute, battery	Shorter rounds	May under-sample difficult users
Utility-aware	Loss, uncertainty, contribution	Faster short-term progress	Selection bias
Diversity-aware	Data/model diversity proxies	Broader coverage	Extra estimation overhead

Client selection changes both system efficiency and which populations influence the global model.

Communication Is a First-Class Resource

In FL, a “cheap” local computation can be preferable if it replaces an expensive model exchange.

Frequency

More local steps reduce how often clients communicate.

Payload

Compression reduces bytes per update.

Model scope

Pruning or partial updates reduce the transmitted parameter set.

Total communication $\approx$ number of rounds $\times$ bytes per round $\times$ participating clients

Reducing one factor may increase another—so total cost matters more than any single number.

Communication-Efficient Techniques

Local-update SGD

Perform multiple local steps before aggregation.

Quantization

Represent updates with fewer bits.

Sparsification

Transmit only a subset of update coordinates.

Error feedback

Accumulate compression residuals for later rounds.

Pruning

Remove low-importance model parameters.

Partial parameter updates

Communicate only selected layers or modules.

Each method reduces a different component of communication cost, but may change convergence behavior.

Quantization vs. Sparsification

Quantization

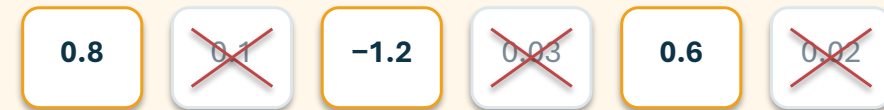
Keep many coordinates; reduce precision per value.



fewer bits per value

Sparsification

Keep high-importance coordinates; omit many others.

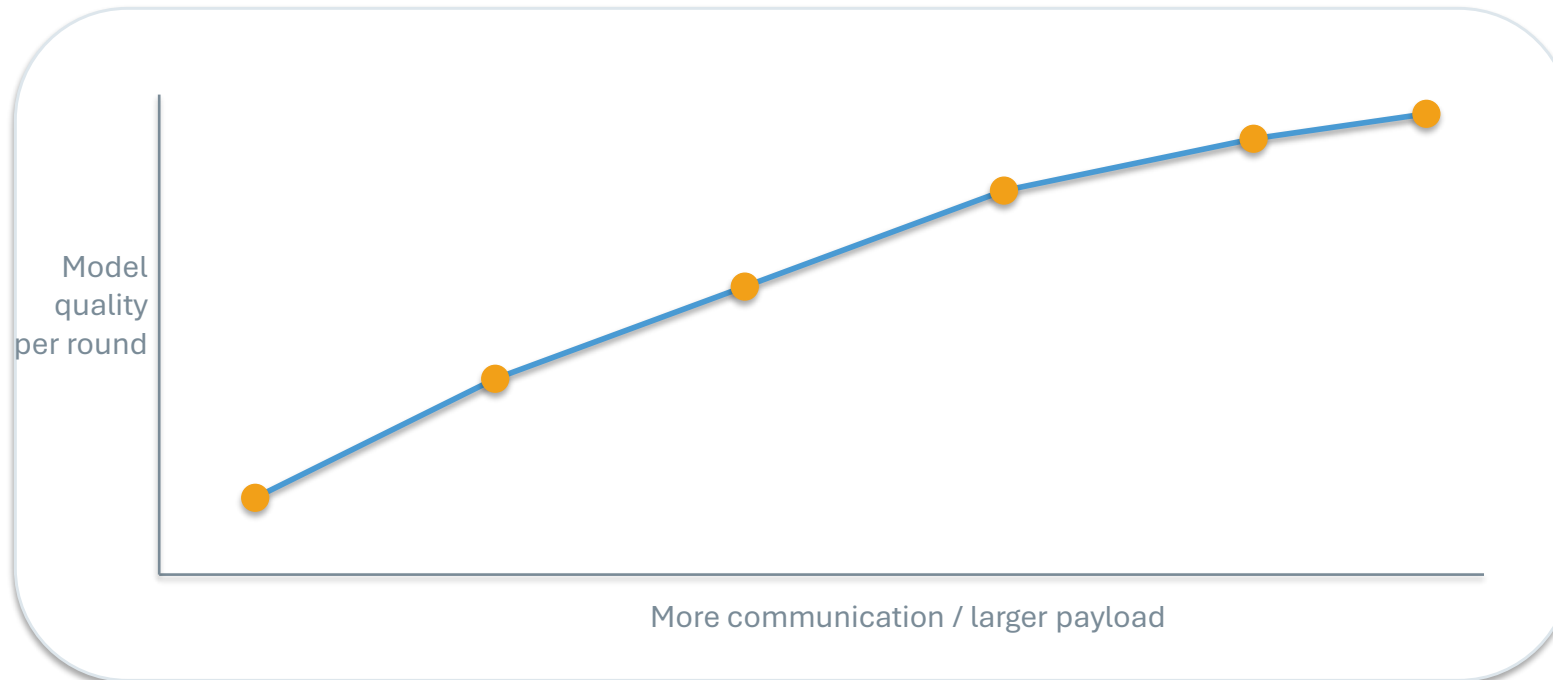


fewer transmitted values

Quantization reduces precision; sparsification reduces the number of transmitted coordinates.

The Communication–Accuracy Trade-off

Communication reduction usually trades against optimization quality.



Better question

How much total communication and wall-clock time are needed to reach a target accuracy?

Minimize end-to-end cost at a target quality—not simply bytes per round.

How to Report Efficiency Correctly

Communication rounds

How often was aggregation performed?

Bytes transferred

Count uplink and downlink explicitly.

Local computation

Report epochs / steps and model size.

Wall-clock time

Include waiting, communication, and client compute.

Energy / memory

Relevant for mobile or edge deployment claims.

Accuracy target

Measure efficiency at comparable quality.

A method is not 'efficient' just because one metric is smaller; report the full cost picture.

In-Class Design Exercise

Scenario

100 smart-home gateways have heterogeneous bandwidth, compute speed, and battery budgets.

Design 1

Choose how many clients participate per round and how they are selected.

Design 2

Choose E local epochs and one communication-reduction technique.

Defend your choice

State what you optimize: accuracy, latency, fairness, energy, or communication.

A strong answer makes the objective, assumptions, and trade-offs explicit.

Key Takeaways

- 1 Real FL systems face both statistical and system heterogeneity.
- 2 Synchronous rounds can be dominated by stragglers, deadlines, and dropout.
- 3 Client selection changes both systems efficiency and which populations influence training.
- 4 Communication cost can be reduced through local computation, quantization, sparsification, pruning, and partial updates.
- 5 Efficiency claims should report quality, communication, computation, and wall-clock cost together.

Mental model: identify the bottleneck → choose the systems mechanism → check the statistical side effects.

Reading & Preparation

Main textbook

Ludwig & Baracaldo (2022)
Review FL systems, scalability, stragglers, and communication constraints.

Supplementary reference

Kairouz et al. (2021)
Focus on systems constraints, communication, and client participation.

Prepare for Week 6 — Personalized Federated Learning and Evaluation

- 1 Think about why one global model may not fit every client.
- 2 Review local fine-tuning and global–local model trade-offs.
- 3 Consider which client-level metrics are needed for fair evaluation.

Bring one example where clients genuinely need different models—not just different training speeds.

Q & A

Thank You.

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